

Exp. No: 4 Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.

Aim: To write program for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB.

Software required:

- SCILAB version 6.0.1

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kihuy.sci (C:\Users\abhis\Documents\kihuy.sci) - SciNotes
File Edit Format Options Window Execute ?
kihuy.sci (C:\Users\abhis\Documents\kihuy.sci) - SciNotes
kihuy.sci X
22
23 // -Volume-charge-density
24 rv = -q / ((4/3) * pi * r^3);
25 disp("The volume-charge-density is (C/m^3):");
26 disp(rv);
27
28 // -Distance-values-for-plotting
29 x = linspace(0, 10, 50);
30
31 // -Exponential-variation
32 y = r1 * exp(-0.1 * x);
33 z = r2 * exp(-0.1 * x);
34 v = rv * exp(-0.1 * x);
35
36 // -Create-figure
37 scf(1);
38
39 // -Line-charge-density-plot
40 subplot(2, 2, 1);
41 plot2d3(x, y);
42 title("Line-Charge-Density");
43 xlabel("Length- (m)");
44 ylabel("rho_l- (C/m)");
45
46 // -Surface-charge-density-plot
47 subplot(2, 2, 2);
48 plot2d3(x, z);
49 title("Surface-Charge-Density");
50 xlabel("Length- (m)");
51 ylabel("rho_s- (C/m^2)");
52
53 // -Volume-charge-density-plot
54 subplot(2, 2, 3);
55 plot2d3(x, v);
56 title("Volume-Charge-Density");
57 xlabel("Length- (m)");
58 ylabel("rho_v- (C/m^3)");
59
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Scilab 2026.1.0 Console

File Edit Control Applications ?

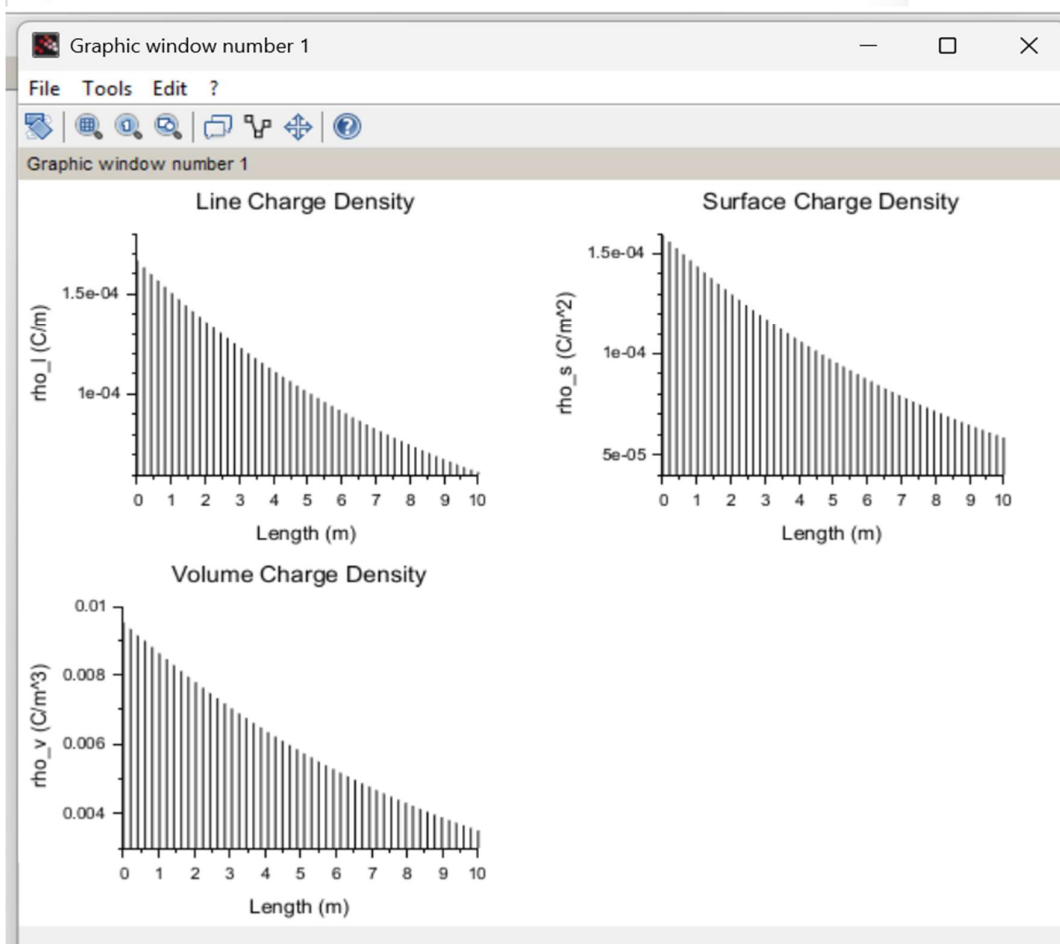
Enter the charge value (C): 5*10^-6

Enter the length value (m): 3*10^-2

Enter the radius value (m): 5*10^-2

"The line charge density is (C/m):"
0.0001667
"The surface charge density is (C/m^2):"
0.0001592
"The volume charge density is (C/m^3):"
0.0095493

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④ Given:-

$$Q = 5 \times 10^{-6} \text{ C}$$

$$z = 3 \times 10^{-2} \text{ m} = 0.03 \text{ m}$$

$$r = 5 \times 10^{-2} \text{ m} = 0.05 \text{ m}$$

Formula

$$P_1 = \frac{Q}{z}$$

$$P_1 = \frac{5 \times 10^{-6}}{3 \times 10^{-2}}$$

$$= \frac{5 \times 10^{-6}}{0.03}$$

$$= 1.6667 \times 10^{-4} \text{ C/m}$$

$$P_1 = 0.0001667 \text{ C/m}$$

⑤ Formula

$$P_2 = \frac{Q}{\frac{4}{3} \pi r^2}$$

$$= \frac{5 \times 10^{-6}}{4 \times 3.1416 \times (0.05)^2}$$

$$= \frac{5 \times 10^{-6}}{4 \times 3.1416 \times 0.0025}$$

$$= \frac{5 \times 10^{-6}}{0.031416}$$

$$= 1.592 \times 10^{-4} \text{ C/m}^2$$

$$P_2 = 0.0001592 \text{ C/m}^2$$

Formula

$$P_0 = \frac{Q}{\frac{4}{3} \pi r^3}$$

$$= \frac{5 \times 10^{-6}}{\frac{4}{3} \times 3.1416 \times (0.05)^3}$$

$$= \frac{5 \times 10^{-6}}{\frac{4}{3} \times 3.1416 \times 0.00125}$$

$$= \frac{5 \times 10^{-6}}{4.1888 \times 0.00125}$$

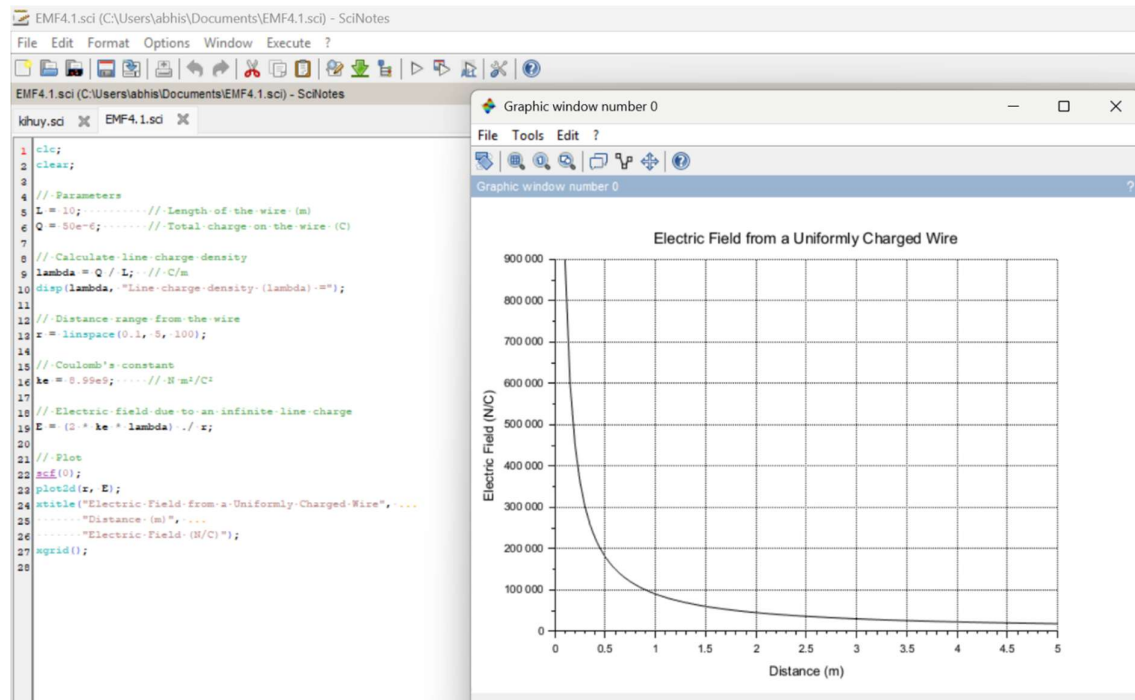
$$= \frac{5 \times 10^{-6}}{0.005236}$$

$$= 9.5493 \times 10^{-3} \text{ C/m}^3$$

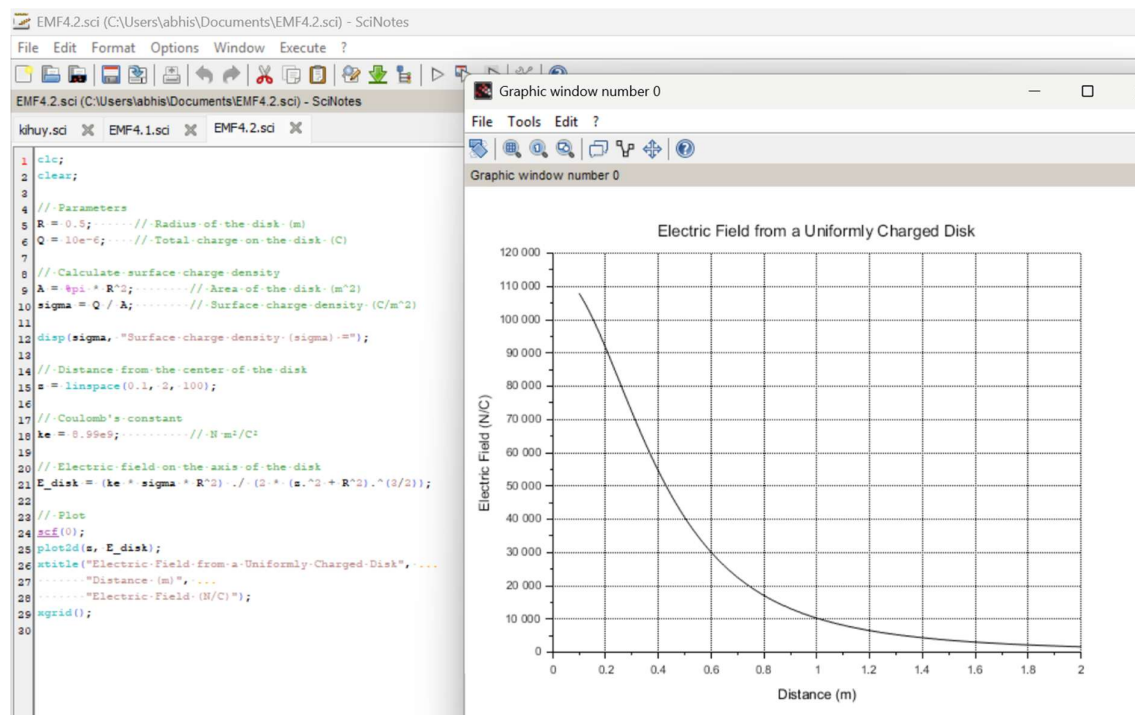
$$= 9.5493 \times 10^{-3} \text{ C/m}^3$$

$$P_0 = 0.0095493 \text{ C/m}^3$$

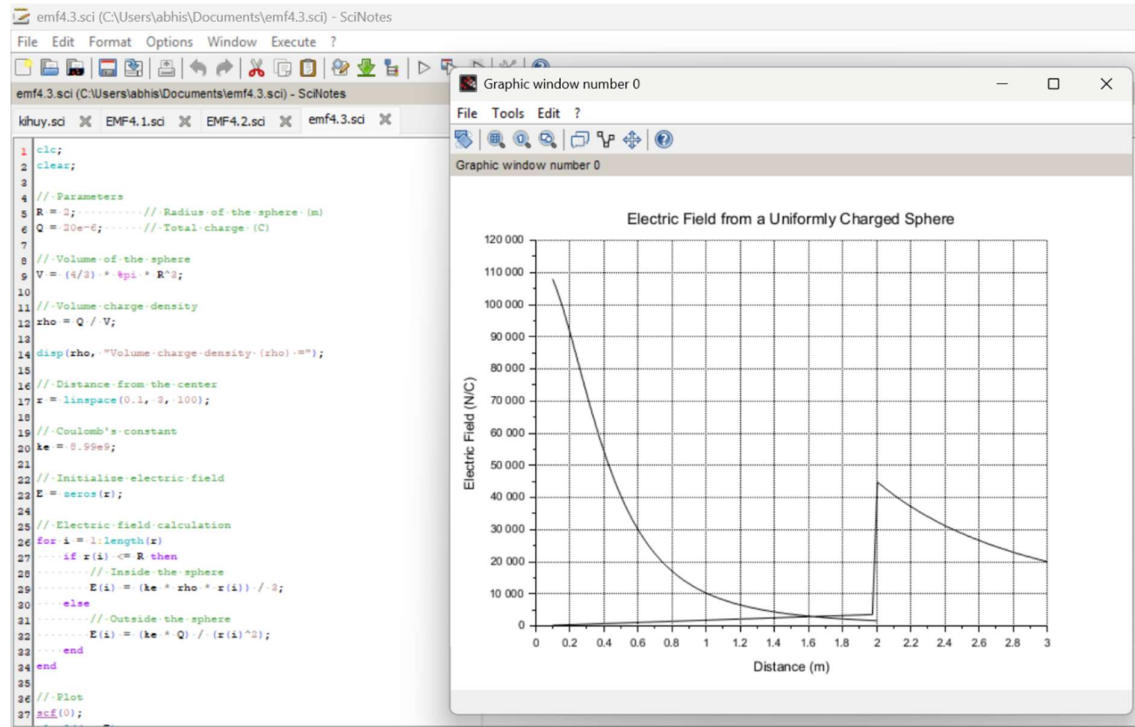
Case 1: A long wire of length $L=10\text{m}$ carries a uniform charge of $Q=50\mu\text{C}$. The goal is to calculate the linecharge density λ and determine how the electric field behaves as a function of distance from the wire.



Case 2: Given a uniformly charged disk of radius $R=0.5\text{m}$ with total charge $Q=10\mu\text{C}$, calculate the surface charge density and plot the electric field along the axis of the disk. Explain the behavior of the electric field as a function of distance from the center of the disk.



Case 3: Given a uniformly charged sphere of radius $R=2\text{m}$ with total charge $Q=20\mu\text{C}$, calculate the volume charge density and plot the electric field both inside and outside the sphere. Explain the difference in electric field behavior for points inside and outside the sphere.



Result : Thus the program was verified for continuous Charge Distribution in electric field. The different types of charge density there are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using SCILAB was successfully.